Homework 3 - 193DS Spring

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```r
# attaching packages

library(tidyverse)
library(janitor)
library(here)
library(effectsize)
library(readxl)
library (ggeffects)
library(rstatix)

# reading in kelp data
kelp <- read.csv(
  here("data", "temp_kelp.csv"))

# reading in personal data
my_data <- read.csv(
  here("data","personal_data.csv"))
```

Problem 1. Giant Kelp Fronds

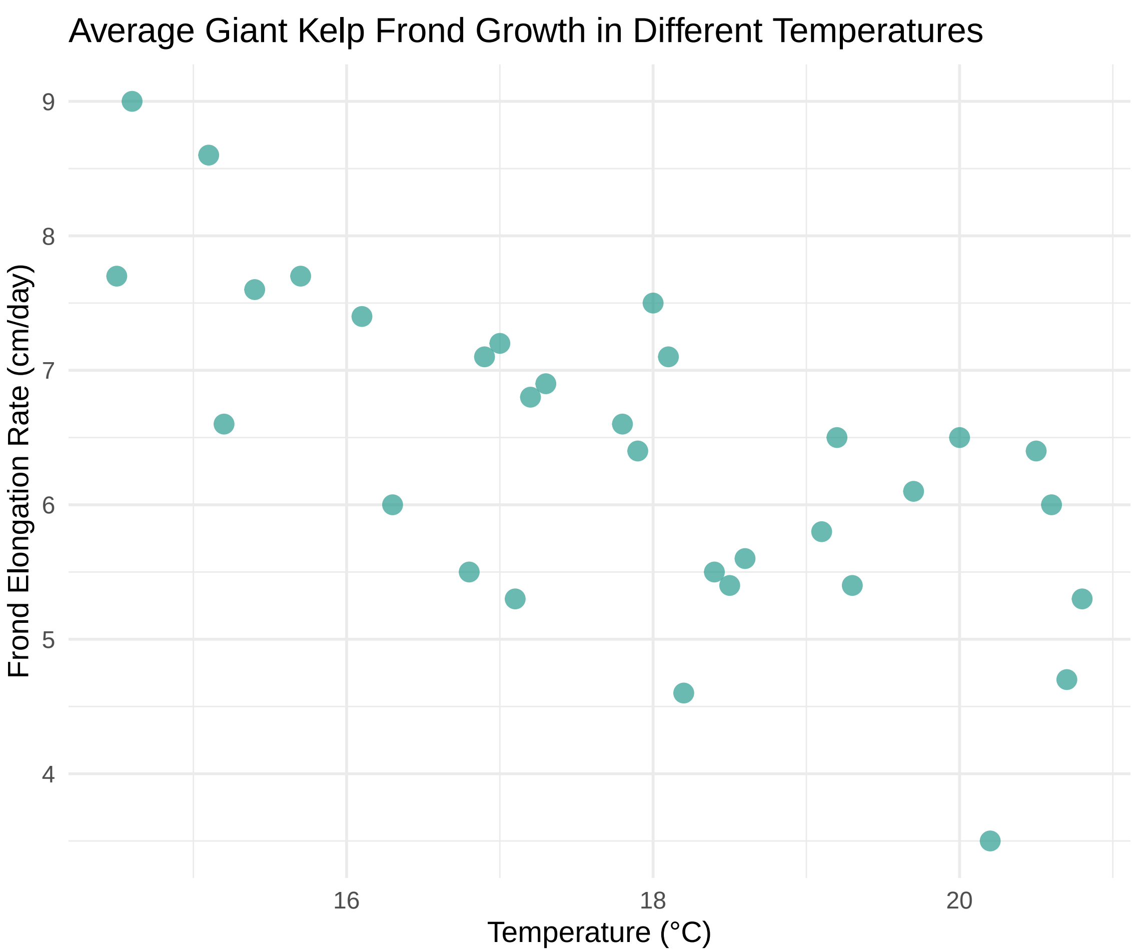
a. An appropraite test

To measure the strength and direction of the relationship between kelp frond length and temperature, I will use two correlation tests: Pearson's Correlation Coefficient and Spearman's Rank Correlation. Pearson's Correlation Coefficient assumes a linear relationship between two continuous variables and provides an r value describing the strength and direction of that relationship. Spearman's Rank Correlation is the non-parametric alternative, which converts values to ranks and exemplifies outliers. Since both tests produce similar results, it can strengthen the conclusion while any disagreement between them may suggest outliers or a non-linear relationship in the data.

b. Create a visualization

```r
# base layer: ggplot
ggplot(data = kelp,
       # x-axis: temperature
       # y-axis: kelp elongation rate (cm/day)
       mapping = aes(x = temp_c,
                     y = kelp_elong)) +
  # first layer: adding data points
```

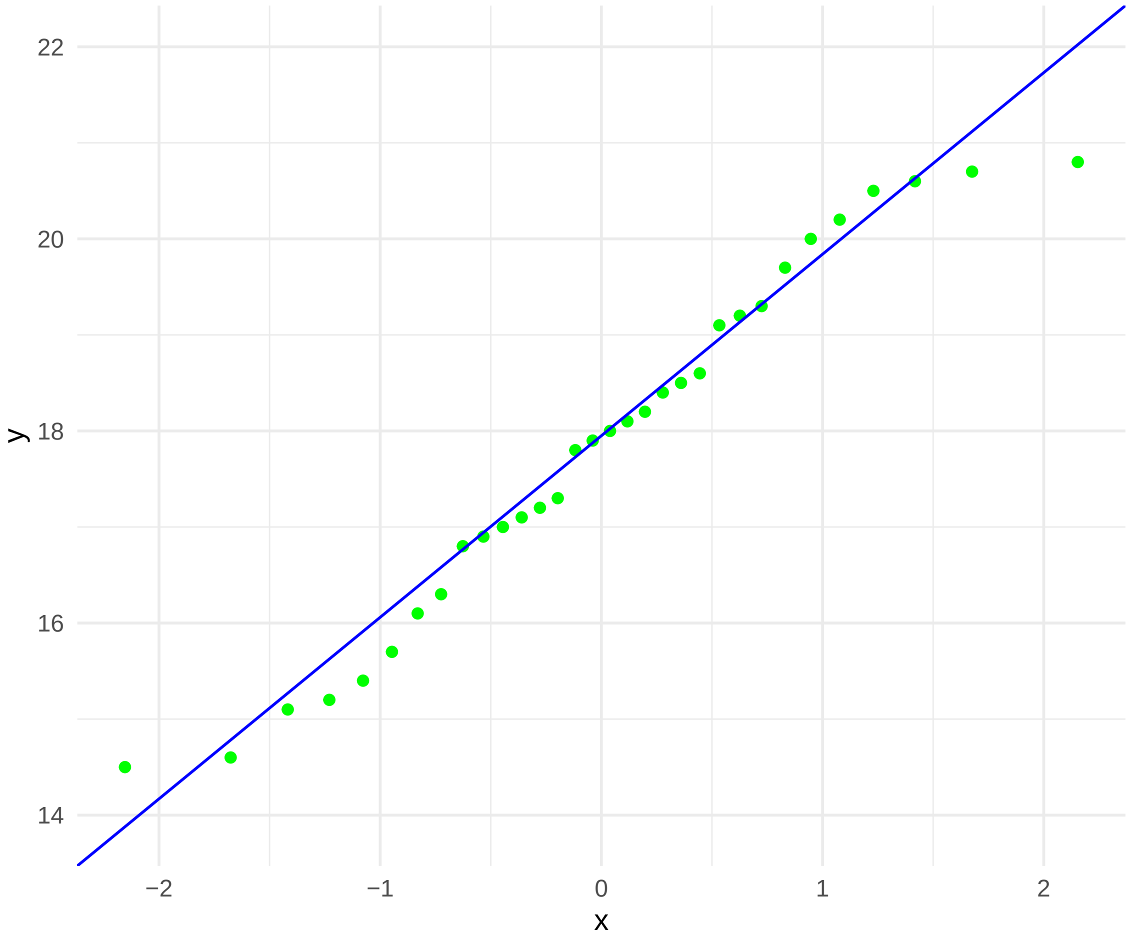
```r
geom_point(color = "#2a9d8f",
           alpha = 0.7,
           size = 3) +
# labeling x- and y-axis
labs(
  x = "Temperature (°C)",
  y = "Frond Elongation Rate (cm/day)",
  title = "Average Giant Kelp Frond Growth in Different Temperatures") +
# adding different theme from default
theme_minimal()
```



c. Checking assumptions and running tests

Checking normality of temperature

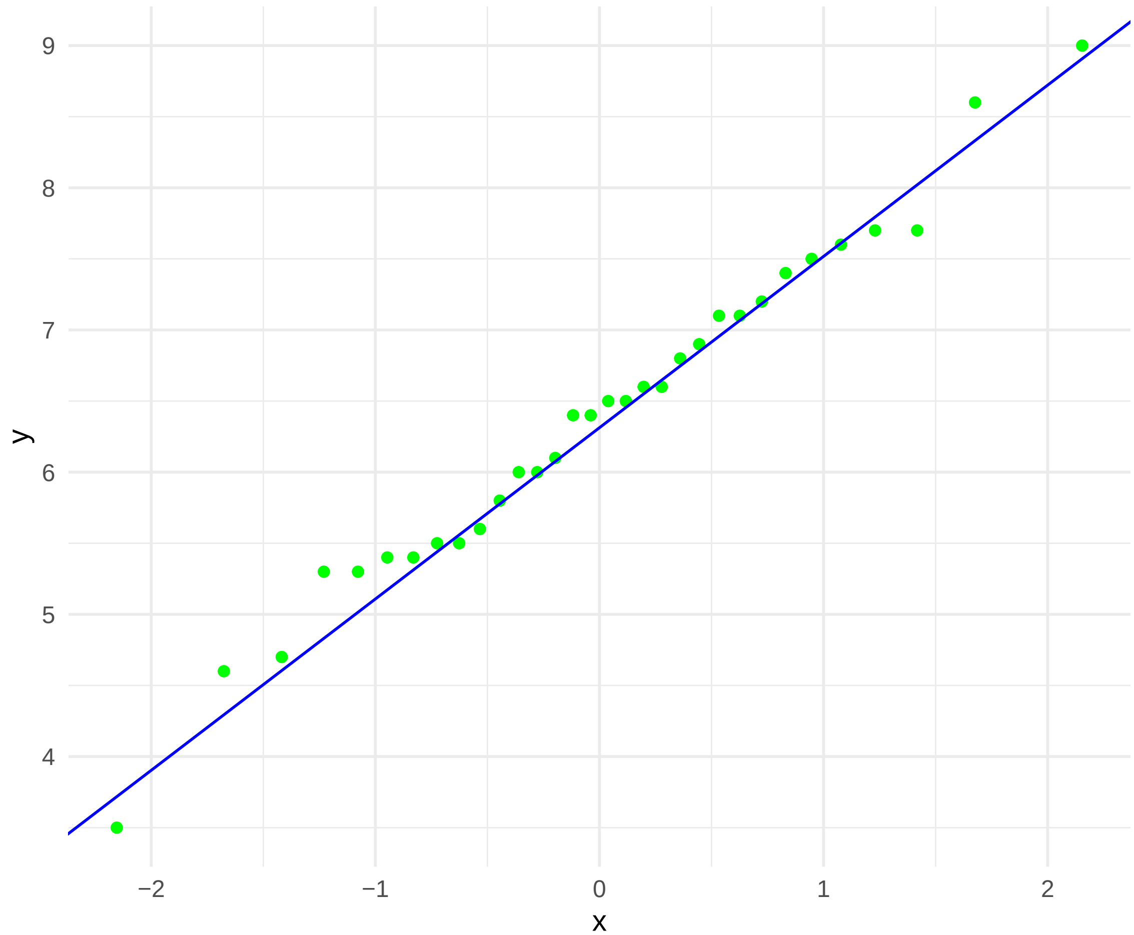
```
# base layer: ggplot
ggplot(data = kelp,
       # setting sample to only temperature
       aes(sample = temp_c)) +
  # first layer: qq plot + points
  geom_qq(color = "green") +
  # second layer: qq line
  geom_qq_line(color = "blue") +
  # taking out legend
  theme(legend.position = "none") +
  # changing theme from default
  theme_minimal()
```



Checking normality of kelp elongation rate

```
# base layer: ggplot
ggplot(data = kelp,
       # setting sample to only kelp elongation rate
       aes(sample = kelp_elong)) +
  # first layer: qq plot + points
  geom_qq(color = "green") +
  # second layer: qq line
  geom_qq_line(color = "blue") +
  # taking out legend
  theme(legend.position = "none") +
```

```r
  # changing theme from default
  theme_minimal()
```



Linearity was assessed visually using a scatterplot of temperature (°C) vs. giant kelp frond elongation rate (cm/day). Normality of each variable was assessed individually using Q-Q plots for both temperature and elongation rate. The scatterplot suggests a negative linear relationship between the two variables, and both Q-Q plots show values falling close to the reference line, indicating normality. All assumptions for Pearson's correlation appear satisfied.

```r
# Pearson's correlation coefficient test
cor.test(kelp$temp_c,
         kelp$kelp_elong,
         method = "pearson")
```

```
	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate I used a Pearson's correlation coefficient. This tests was an appropriate choice because both variables are continuous and the scatterplot suggested a linear relationship between them.

Temperature was a significant negative predictor of giant kelp frond elongation rate ($r = -0.69$, $p < 0.001$, $R^2 = 0.473$). As seawater temperature increased, kelp fronds grew more slowly, suggesting that warmer water conditions are affect Macrocystis pyrifera and inhibit its growth. This pattern is consistent with what we know about kelp ecology, giant kelp thrives in cold, nutrient-rich water, and elevated temperatures likely symbolize conditions such as reduced upwelling that limit the nutrients available for growth.

e. Test implications

Our analysis found a strong negative relationship between seawater temperature and giant kelp frond elongation rate, meaning kelp grows significantly slower as water temperatures rise. This is very concerning given ongoing ocean warming trends. If temperatures continue to increase, we should expect noticeable declines in kelp growth rates, which could have cascading effects on the broader kelp forest ecosystem. I'd recommend to prioritize monitoring sites where temperatures are already at the higher end of the range we observed, as those are likely the most at-risk locations for growth suppression.

f. Double check my own work

```r
# Spearman's Rank Correlation test
cor.test(kelp$temp_c,
         kelp$kelp_elong,
         method = "spearman")
```

```
	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

Both Pearson's correlation coefficient and Spearman's Rank Correlation were used to assess the strength and direction of the relationship between water temperature (°C) and giant kelp frond elongation rate (cm/day). Pearson's correlation coefficient ($r = -0.688$, $t(30) = -5.189$, $p < 0.001$) showed a moderately negative linear relationship between the two variables, meaning that as water temperature increased, giant kelp frond elongation rate significantly decreased. The t-statistic of -5.189 confirmed that this correlation was significantly different from zero. Spearman's Rank Correlation ($\rho = -0.689$, $S = 9216.1$, $p < 0.001$) obtained this finding by ranking both temperature and elongation rate values and measuring the monotonic relationship between them. It confirms that higher ranked temperatures consistently corresponded with lower ranked elongation rates. The almost identical correlation coefficients produced by both tests ($r = -0.688$, $\rho = -0.689$) strengthen confidence in the result. The agreement between a parametric and non-parametric tests suggests the relationship is not being driven by outliers or violations of normality.

Problem 2. Personal data

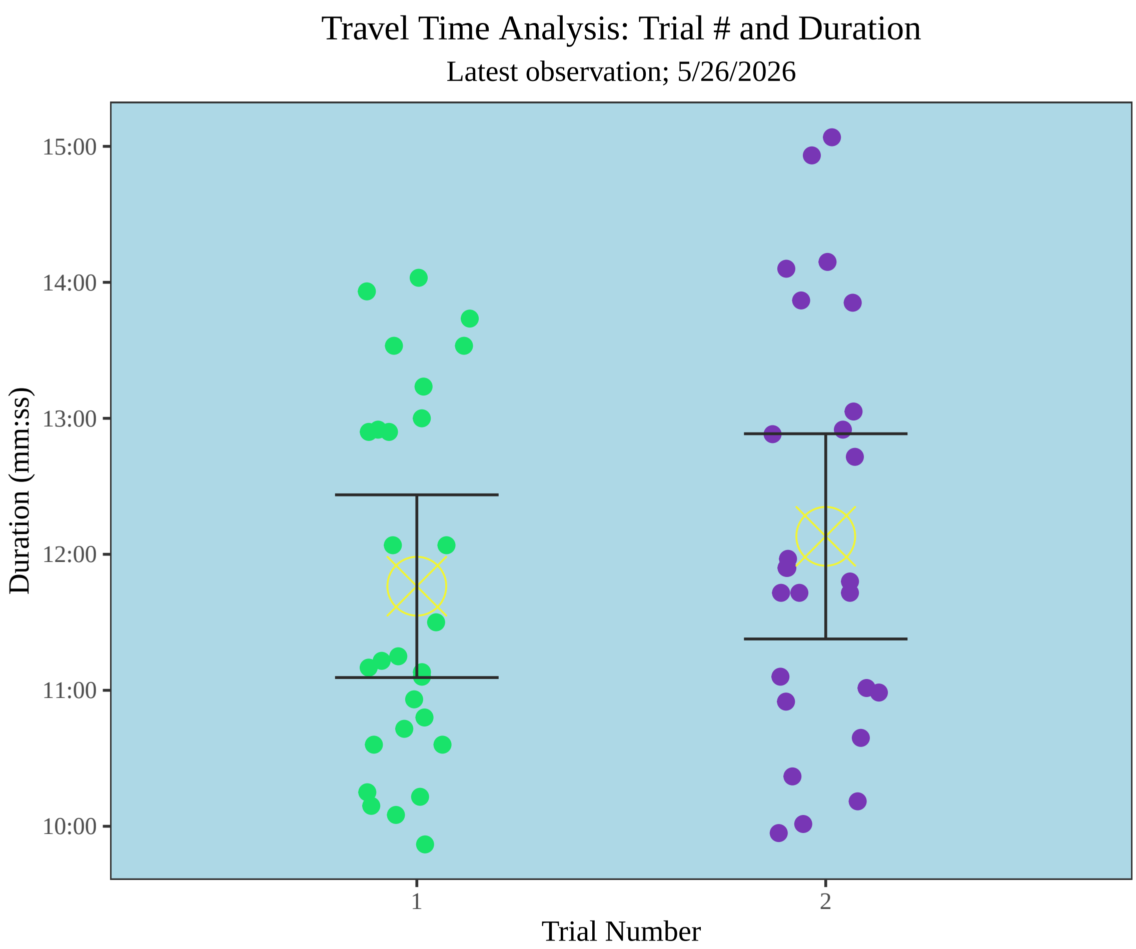
a. Updating visualizations

```r
# creating new object called my_data_clean
my_data_clean <- my_data |>
# cleaning column names so only lowercase and underscores are used
clean_names() |>
  mutate(duration_sec = as.numeric(ms(duration_mm_ss)))
```

```r
# define mm:ss label format
mmss_format <- function(x) sprintf("%d:%02d", x %/% 60, x %% 60)
# base layer: gpplot
ggplot(data = my_data_clean,
       # x-axis: trial number
       # y-axis: duration
```

```
     mapping = aes(x = factor(trial_number),
                   y = duration_sec,
                   color = factor(trial_number))) +
  # first-layer: jitter horizontally to organize visualization
  geom_jitter(width = 0.13,
              height = 0,
              size = 2.5) +
  # mean points on graph
  stat_summary(fun = mean, # calculate average
               geom = "point",  # display as point
               color = "#f2f533", # change color of point
               shape = 13, # change shape of point
               size = 10) +
  # standard deviation error bars
  stat_summary(fun.data = mean_sdl, , # calculate standard deviation
               fun.args = list(mult = 0.5), # tighter view of variability
               geom = "errorbar", # adding error-bar
               width = 0.4, # changing width
               color = "#2b2b2b") + # changing color
  # assigning colors to trial 1 and trial 2
  scale_color_manual(values = c("1" = "#19e36a",
                                "2" = "#7836b5")) +
  # assigning a new and organized scale for y-axis
  scale_y_continuous(
    breaks = seq(600, max(my_data_clean$duration_sec, na.rm = TRUE), by = 60),
    minor_breaks = seq(600, max(my_data_clean$duration_sec, na.rm = TRUE), by = 30),
    labels = mmss_format
  ) +
  # adjusting plot (decreasing space in between categories and plot borders)
  scale_x_discrete(expand = expansion(add = 0.7, mult = 0.05)) +
  # changing theme from default
  theme_bw(base_family = "serif") +
   # changing panel and plot background
  theme(
    panel.background = element_rect(fill = "lightblue"),
    plot.background = element_rect(fill = "transparent", color = NA)
  ) +
  # taking out the minor grid
  theme(panel.grid.minor = element_blank()) +
  # taking out major grid
  theme(panel.grid.major = element_blank()) +
  # taking out legend
```

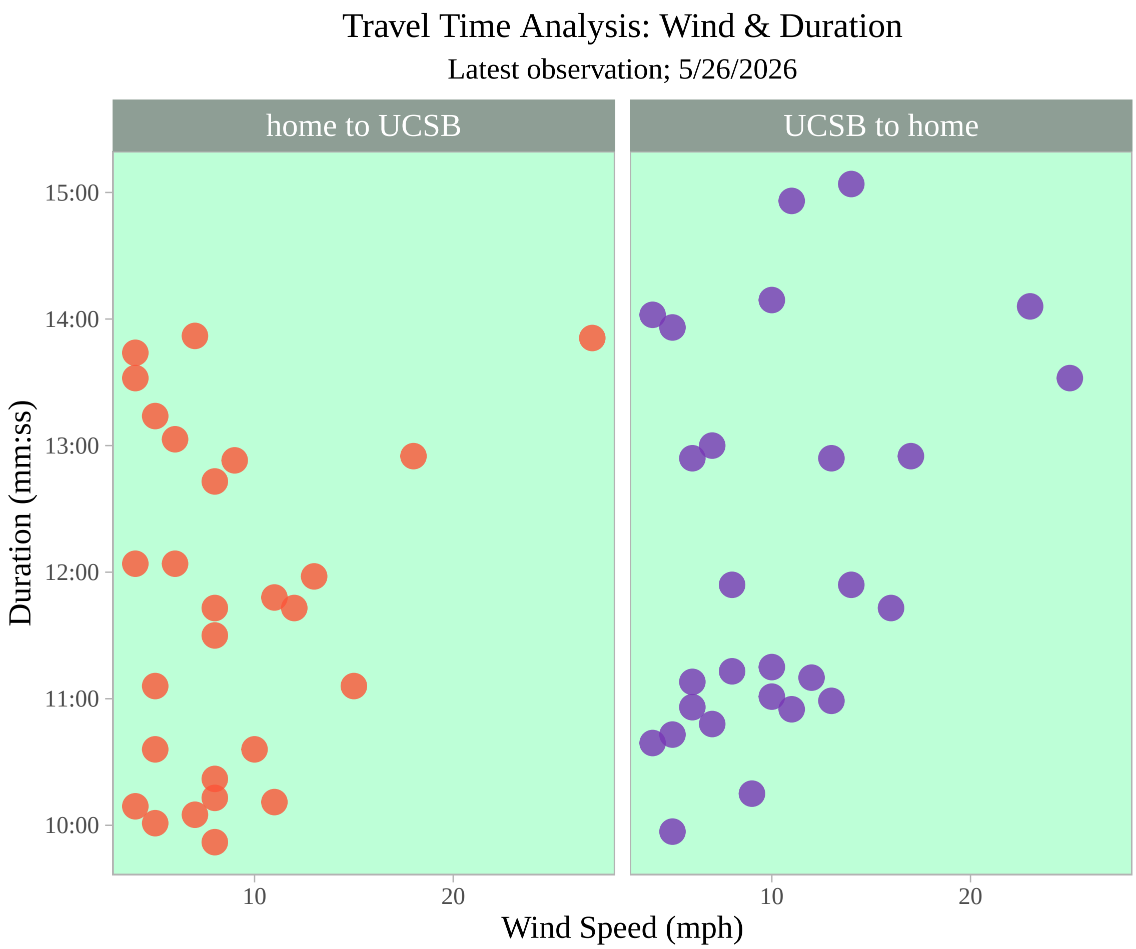
```
  theme(legend.position = "none") +
  # adjusting title position
  theme(plot.title = element_text(hjust = 0.5)) +
  # adjusting subtitle position
  theme(plot.subtitle = element_text(hjust = 0.5)) +
  # adding title and labeling to x- and y-axis
  labs(x = "Trial Number",
       y = "Duration (mm:ss)",
       title = "Travel Time Analysis: Trial # and Duration",
       subtitle = "Latest observation; 5/26/2026")
```




```
# base layer ggplot
ggplot(data = my_data_clean,
       # x-axis = wind speed
```

```
     # # y-axis = direction of trip
     # # color varies by direction of trip
    aes(x = wind_speed_mph,
                  y = duration_sec,
                  color = direction_of_trip)) +
# adding points
geom_point(size = 4, alpha = 0.8) +
# splitting into two graphs (one per category)
facet_wrap(~ direction_of_trip) +
# assigning a new and organized scale for the y-axis
scale_y_continuous(
  breaks = seq(600, max(my_data_clean$duration_sec, na.rm = TRUE), by = 60),
  minor_breaks = seq(600, max(my_data_clean$duration_sec,
                              na.rm = TRUE), by = 30),
  labels = mmss_format
) +
 # assigning colors for each categories
scale_color_manual(values = c("home to UCSB" = "#fc5538",
                              "UCSB to home" = "#7836b5")) +
 # changing theme from default
theme_light(base_family = "serif") +
# changing panel and plot background
theme(
  panel.background = element_rect(fill = "#bdffd7"),
  plot.background = element_rect(fill = "transparent", color = NA)
) +
# labeling x- and y-axis and adding title
labs(title = "Travel Time Analysis: Wind & Duration",
     subtitle = "Latest observation; 5/26/2026",
     x = "Wind Speed (mph)",
     y = "Duration (mm:ss)") +
# taking out the minor grid
theme(panel.grid.minor = element_blank()) +
# taking out major grid
theme(panel.grid.major = element_blank()) +
# taking out legend
theme(legend.position = "none") +
# adjusting title position
theme(plot.title = element_text(hjust = 0.5)) +
# adjusting subtitle position
theme(plot.subtitle = element_text(hjust = 0.5)) +
# adjusting facet_wrap text and color
```

```
theme(
strip.text = element_text(size = 12, color = "white"),
strip.background = element_rect(fill = "#8e9e95")) +
# adjusting x- and y-axis text
theme(axis.title = element_text(size = 12))
```



b. Captions

Figure 1. Scatter plot with mean travel duration for trial number 1 and 2 and error bars displaying $\pm$ 0.5 standard deviations around the mean. Individual observation points in each trial are denoted by x-axis (1,2) as well as colors (trial 1: green, trial 2: purple). Trial number averages are denoted by the yellow circles with X's going through. Error bars display the 95% confidence interval around each regional mean. Data source: Michael Torbensen. 2026. Average bike duration spreadsheet.

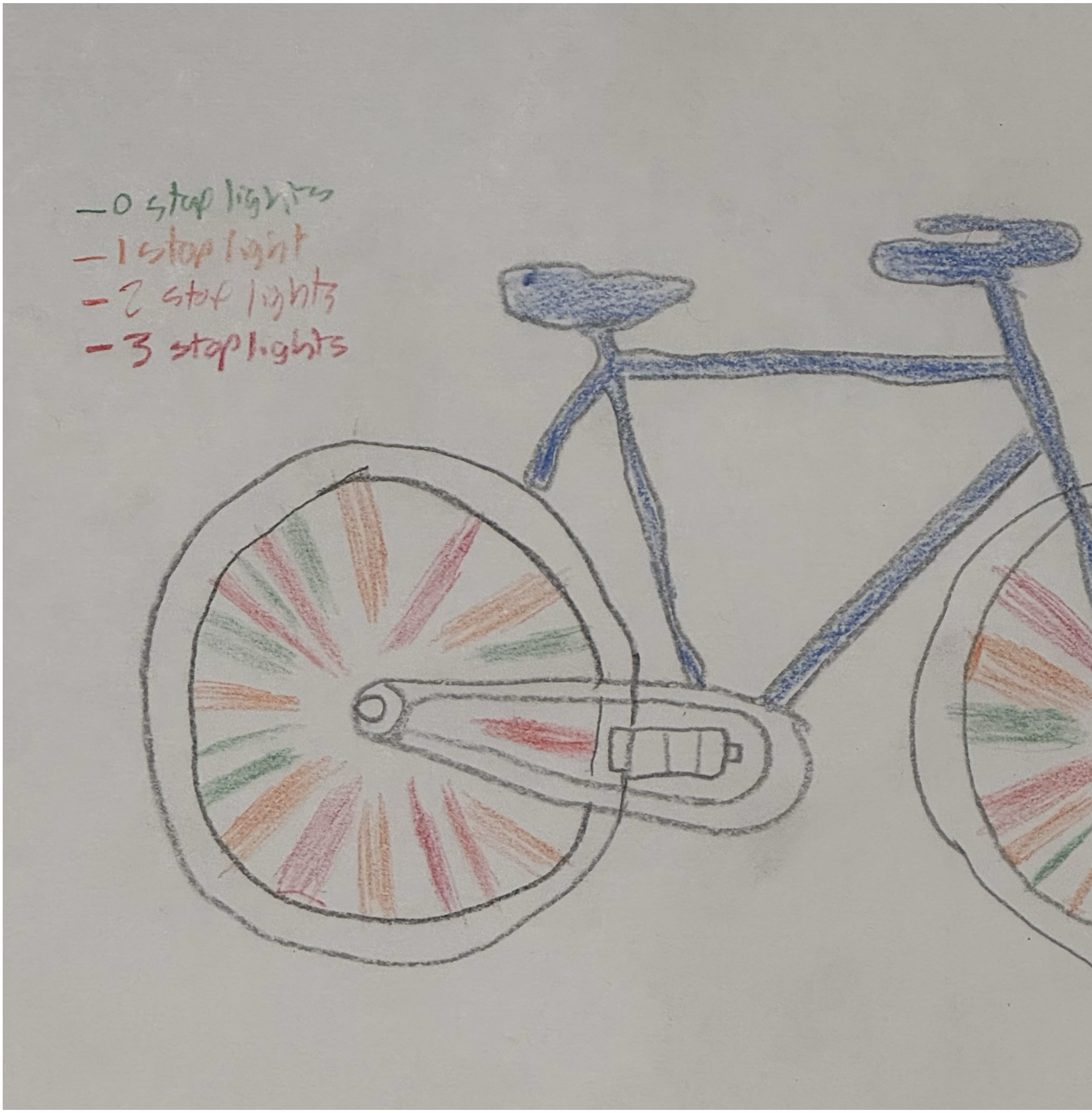
Figure 2. Two separate panels for travel direction show scatter plots with a regression line modeling the average relationship between duration (y-axis) and wind speed (x-axis). The first panel (home to UCSB) features red points indicating individual observations as well as a red regression line. The second panel (UCSB to home) features purple points indicating observations as well as a purple regression line. Data source: Michael Torbensen. 2026. Average bike duration spreadsheet.

Problem 3. Affective visualization

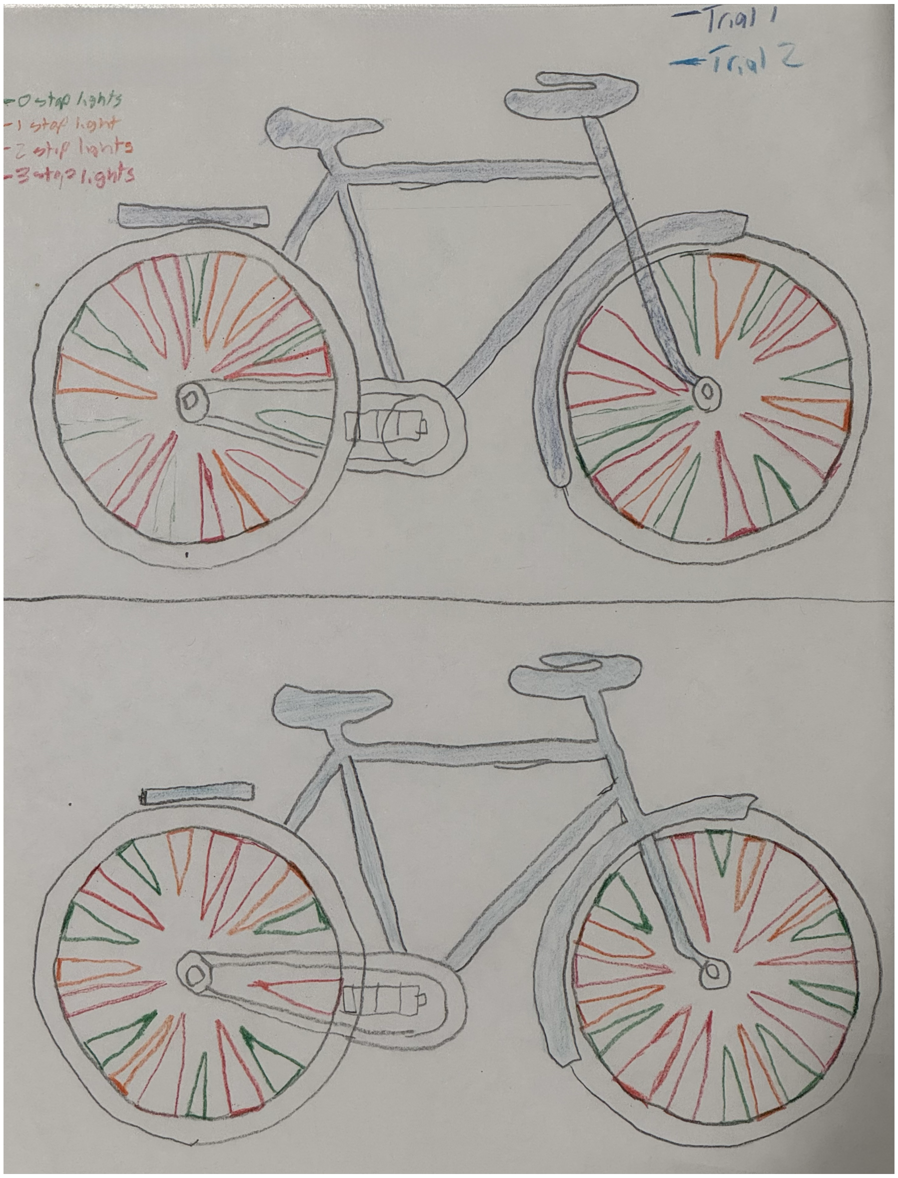
a. Idea for affective visualization

For my affective visualization, I was thinking of representing mostly all of my data with two bikes using the wheels as a place to represent my individual observations. The separate bikes will represent all data points that fall on either trial number one or two (categorical data). The rear wheel will represent home to UCSB data observations and the front wheel will represent UCSB to home observations (categorical data). Within the tires, each spoke will represent wind speed (width of spoke), travel duration (length of spoke), and number of stop lights (color of spoke). Also, I'm thinking of adding a sheen/gloss to the spoke with the quickest trip and the slowest trip will be a little transparent.

b. Sketch for visualization



c. Draft of visualization



d. Artist statement

I enjoy getting to my destinations as fast as possible and learning what variables makes me slower and what variables makes me faster. So, I have created a two panel plot that expresses two different categorical variables and 3 different continuous variables. I will be using colored pencils on a blank sheet of white paper to create my final drafted drawing.

e. Presentation

View Presentation

Problem 4. Statistical critique

a. Revisit and summarize

In this paper, the author is wondering how effective re-connected floodplains are at de-nitrifying bodies of water that are useful to humans and other biological communities. Out of the tests conducted, analysis of variance (ANOVA) is the main test that the author used.

				NO_3^-					
treatment	**time period**	**Site**	**GW/SW**	**N**	**mean**	**SE**	***df***	***F***	***p***
restored	pre-restoration	CVP	GW	128	1.56	0.08	1,138	0.04	0.84
			SW	15	1.61	0.03			
restored	post-restoration	CVP	GW	308[*]	1.33	0.06	1,340	2.63	0.11
			SW	43	1.08	0.08			
control	pre-restoration	Intervale	GW	27	0.84	0.06	1,30	0.09	0.77
			SW	8	0.88	0.10			
control	post-restoration	Intervale	GW	168	0.97	0.04	1,187	2.93	0.09
			SW	30	1.12	0.06			

b. Visual Clarity

The table clearly shows the results of the ANOVA tests by including the key statistical components needed for interpretation: degrees of freedom, F-statistic, and p-value. The inclusion of N (sample size), mean, and SE (standard error) alongside the test statistics is also useful because it gives the reader context for the raw data underlying each comparison, rather than just isolating the test results.

c. Aesthetic clarity

The authors handled visual clutter in a way that made the graph legible but not with outstanding quality. The categories listed in the table are bold so that each separate category cannot be confused with the other. Although, all data have the same font style, size, and type. So, if there is any data that is particularly useful for the validity of the test results, it cannot be deduced by looking at the graph. That being said, the organization by treatment, time period, and site makes it easy to compare across all four ANOVAs.

d. Recommendations

Typically, for an ANOVA test, it's common to mention the sum of squares (SS) and mean squares (MS) to measure the among-group variation and within-group variation as this is where the F-statistic and effect size come from. This would give more information to the reader and increase clarity and comprehension of the data. Also, since the group sizes are so large, the SE numbers are very small. If the SE column was replaced with SD, the spread of the nitrate measurements in each group would be much more visible and one can judge whether differences between group means are biologically significant.